

KRYVYI RIH, Ukraine

WMO: 337910

Lat: 48.0536N

Lon: 33.2111E

Elev: 407

StdP: 14.48

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.3	5.2	-5.6	4.1	0.3	0.0	5.6	6.1	26.7	23.5	24.5	27.6	8.5	0	0.580

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.2	92.9	68.6	89.5	67.7	86.2	66.9	71.7	85.6	70.1	83.6	68.6	81.8	9.2	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.5	102.6	77.3	65.8	96.8	75.3	64.3	91.5	73.8	35.7	86.1	34.3	83.9	33.1	81.7	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	20.1	17.8	-6.4	96.9	6.6	3.4	-11.2	99.3	-15.0	101.3	-18.8	103.1	-23.6	105.6
			-6.9	75.4	6.4	2.6	-11.5	77.3	-15.3	78.8	-18.9	80.3	-23.6	82.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.6	25.3	27.7	37.1	50.8	61.5	68.4	72.5	71.7	61.5	49.1	38.2	29.4
	DBStd	18.71	10.29	10.20	8.26	7.24	7.40	6.22	5.61	6.52	7.39	8.30	8.70	9.75
	HDD50	3001	766	624	404	76	6	0	0	0	6	118	363	638
	HDD65	6334	1231	1044	866	429	157	36	6	14	150	495	803	1103
	CDD50	2839	0	0	3	99	362	552	699	671	351	91	11	0
	CDD65	697	0	0	0	2	49	138	240	220	45	3	0	0
	CDH74	7211	0	0	0	32	541	1356	2429	2358	474	21	0	0
	CDH80	2738	0	0	0	6	145	457	975	1016	137	2	0	0
Wind	WSAvg	8.9	9.8	9.8	10.3	9.6	8.4	8.1	7.7	8.1	8.1	8.4	9.2	9.7
Precipitation	PrecAvg	19.4	1.5	1.3	1.4	1.3	2.0	2.4	2.4	1.3	1.7	1.5	1.5	1.4
	PrecMax	31.5	3.3	3.6	2.5	3.6	3.4	3.7	4.5	3.3	4.8	4.4	3.2	3.2
	PrecMin	10.6	0.3	0.3	0.4	0.0	0.5	0.8	0.6	0.4	0.0	0.0	0.1	0.2
	PrecStd	4.3	0.8	0.9	0.7	1.0	0.9	0.9	1.1	0.7	1.3	1.1	1.0	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.7	51.5	64.1	78.1	89.2	91.7	96.2	98.5	89.4	76.7	62.8	52.1
		MCWB	44.4	46.2	50.0	58.2	65.6	68.3	69.7	68.5	65.3	60.7	56.2	49.3
	2%	DB	43.0	46.3	57.9	71.9	84.0	88.1	92.4	93.5	84.0	70.8	57.3	48.3
		MCWB	41.9	42.4	47.5	55.2	63.6	67.6	69.3	68.2	63.0	58.6	53.2	46.7
	5%	DB	40.2	42.6	53.6	68.1	80.3	85.6	89.2	89.7	79.2	65.7	53.4	44.7
		MCWB	39.1	40.3	44.8	53.7	62.4	67.2	68.4	67.4	61.2	57.0	50.7	43.5
	10%	DB	37.1	39.4	49.2	64.4	76.0	82.2	85.8	85.9	75.2	61.1	50.1	41.8
		MCWB	36.1	37.3	42.8	51.9	61.0	65.7	67.8	66.2	59.5	54.7	47.9	40.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.0	47.5	51.7	60.4	68.0	72.9	74.1	73.6	68.3	64.3	57.4	50.3
		MCDB	45.9	50.4	61.6	73.7	82.3	85.6	87.0	89.6	82.0	72.9	61.9	51.5
	2%	WB	42.0	43.3	48.5	57.2	65.6	70.6	72.0	70.9	65.5	60.8	53.9	46.7
		MCDB	42.7	45.3	56.5	69.2	79.2	82.9	85.6	85.9	77.9	66.2	57.0	48.1
	5%	WB	39.0	40.3	45.9	55.0	63.9	68.8	70.3	69.2	63.4	58.0	51.3	43.5
		MCDB	39.8	42.1	51.4	65.0	76.3	80.9	83.8	83.6	74.2	64.2	53.3	44.4
	10%	WB	35.9	37.5	43.4	53.0	62.1	67.0	68.9	67.5	61.4	55.6	48.2	40.8
		MCDB	36.7	39.0	48.0	62.4	73.8	78.6	82.0	81.6	71.6	61.0	49.8	41.6
Mean Daily Temperature Range	5% DB	MDBR	9.5	11.0	14.7	19.6	20.8	20.5	21.2	21.6	19.7	16.3	10.7	9.1
		MCDBR	9.7	14.1	23.4	25.4	25.7	24.9	25.2	26.1	26.0	21.6	15.4	11.3
		MCWBR	8.7	10.7	14.1	12.6	11.4	10.3	9.1	9.8	11.6	12.0	11.2	9.8
	5% WB	MCDBR	9.3	12.9	20.2	22.3	22.7	21.7	22.2	23.0	21.8	18.9	13.6	10.6
MCWBR		8.6	10.3	13.3	11.7	11.0	10.2	9.3	9.8	10.9	12.1	10.7	9.5	
Clear-Sky Solar Irradiance	taub	0.327	0.357	0.386	0.445	0.416	0.411	0.440	0.434	0.407	0.370	0.353	0.330	
	taud	2.254	2.206	2.230	2.100	2.270	2.302	2.228	2.236	2.282	2.368	2.329	2.315	
	Ebn at Noon	228	247	260	255	269	271	260	256	251	242	218	212	
	Edh at Noon	25	32	37	47	41	40	43	40	35	28	23	21	
All-Sky Solar Radiation	RadAvg	306	578	963	1458	1856	1983	1968	1772	1229	745	323	223	
	RadStd	61	101	141	133	157	146	108	101	157	101	54	44	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.28	N/A	N/A	+2.23	N/A	N/A	N/A	-281	+252	+180
(47)	Regional (10 neighbors)	+1.35	N/A	N/A	+1.98	N/A	N/A	-247	-333	+216	+144

Nomenclature: See separate page